

TOAST Stroke Classification

Trial of Org 10172 in Acute Stroke Treatment (TOAST) diagnostic criteria for ischemic stroke etiology.

ACUTE ISCHEMIC STROKE

Large Artery (LAA)

Small Vessel (SVO)

Cardioembolic (CE)

Other Det. (ODE)

Undetermined Etiology

1. Large-Artery Atherosclerosis (LAA)

- **Clinical:** Cortical signs (aphasia, neglect, gaze deviation) or brainstem/cerebellar syndrome.
- **Imaging:** Cortical or subcortical/cerebellar/brainstem infarct matching the vascular territory.
- **Vascular:** > 50% stenosis or occlusion of the relevant major extracranial (carotid, vertebral) or intracranial artery.
- **Exclusion:** Must exclude a high-risk cardioembolic source.

2. Small-Vessel Occlusion (SVO / Lacune)

- **Clinical:** Classic lacunar syndrome (pure motor, pure sensory, sensorimotor, ataxic hemiparesis, clumsy hand) **WITHOUT** cortical signs.
- **Imaging:** Normal scan or deep subcortical/brainstem lesion ≤ 2.0 cm.
- **Vascular/Cardiac:** Relevant artery must lack >50% stenosis, and patient must lack high-risk cardioembolic sources.

4. Other Determined Etiology (ODE)

- **Clinical/Imaging:** Infarction of any size with diagnostic proof of a rare/specific underlying mechanism:
- Arterial dissection (e.g. carotid or vertebral dissection)
- CNS vasculitis or systemic vasculopathy
- RCVS (Reversible Cerebral Vasoconstriction Syndrome)
- Moya-Moya disease, CADASIL, or Fibromuscular Displasia
- Prothrombotic/hypercoagulable state (APLS, cancer, DIC)

3. Cardioembolism (CE)

HIGH-RISK SOURCES

- Atrial Fibrillation / Flutter
- Mechanical prosthetic valve
- Left atrial / LAA thrombus
- Recent anterior MI (<3 mo)
- Dilated cardiomyopathy (EF<30%)
- Infective endocarditis
- Sick sinus syndrome / LA myxoma

MEDIUM-RISK SOURCES

- PFO + Atrial Septal Aneurysm
- Mitral valve prolapse
- Mitral annulus calcification
- Bioprosthetic heart valve
- Calcific aortic stenosis
- LV dysfunction (EF 30–40%)
- LA spontaneous echo contrast

5. Undetermined Etiology

- **Due to competing risks:** ≥ 2 potential etiologies found (e.g., active AFib AND $\geq 50\%$ ipsilateral carotid stenosis).
- **Negative evaluation:** Complete diagnostic workup identifies no clear source (Cryptogenic stroke).
- **Incomplete evaluation:** Workup is unfinished (e.g., patient discharged/AMA before Echo or vascular imaging).

ESUS Criteria: non-lacunar stroke, no relevant >50% stenosis, no high-risk cardioembolic source, negative ECG/telemetry ≥ 24 hours.

REQUIRED DIAGNOSTIC WORKUP TO COMPLETE TOAST CLASSIFICATION

- | | |
|---|---|
| ☒ Parenchymal: MRI Brain (DWI/ADC) preferred, or CT Head. | ☒ Vascular: CTA or MRA Head & Neck (or Carotid Duplex + TCD). |
| ☒ Rhythm: EKG + Continuous Telemetry ≥ 24 h (or loop recorder). | ☒ Cardiac: TTE required; consider TEE if cryptogenic / ESUS suspected. |

Original Study: Adams HP Jr, et al. TOAST. *Stroke*. 1993;24:35-41. | **AHA/ASA Guideline:** Kleindorfer DO, et al. 2021 Stroke Prevention. *Stroke*. 2021;52:e364-e467.